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EcoTown Health Tech Internship Assignment: Biomarker Time Series Visualization Dashboard

Assignment Overview

Duration: As needed (no more than 1 week . communicate if more time is needed)

Objective: Extract biomarker data from health reports and create an interactive time series visualization dashboard for lipid profile, creatinine, vitamin D, B12, and HbA1c levels.

This assignment tests your ability to:

- Extract structured data from health reports
- Build interactive time series visualizations
- Apply healthcare data standards and clinical ranges
- Deploy a functional dashboard online

Part 1: Data Extraction & Processing

Step 1.1: Access Source Data

1. **Download reports** from the provided Google Drive folder:
 - URL: <https://drive.google.com/drive/folders/15eJ1ZcLE1PS1fOLJZN1xI0PS5i-KcLNx>
 - Focus on any 2 PDF reports or structured data files in the "Sample Reports" subfolder

Step 1.2: Extract Biomarker Data

Create a Python script to extract the following biomarkers:

Required Biomarkers & Clinical Ranges:

- **Lipid Profile:**
 - Total Cholesterol:



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- LDL:
- HDL:
- Triglycerides:
- **Creatinine:**
- **Vitamin D:**
- **Vitamin B12:**
- **HbA1c:**

Part 2: Time Series Visualization Development

Step 2.1: Choose Development Environment

Select ONE of the following approaches:

Option A: Cursor + V0 + Replit (Recommended)

1. **Cursor IDE:** <https://cursor.sh/>
 - AI-powered code editor
 - Use for Python data processing and JavaScript visualization
2. **V0 by Vercel:** <https://v0.dev/>
 - Generate UI components with natural language
 - Prompt: "Create a healthcare dashboard with time series charts for biomarkers"
3. **Replit:** <https://replit.com/>
 - Host and deploy your dashboard
 - Real-time collaboration features



Option B: Traditional Stack

- Python + Flask + Chart.js
- Host on Heroku free tier

Step 2.2: Visualization Implementation

Create interactive time series charts using any libraries, refer to drive link to access the function health screen recording and screenshots.

Required Visualization Features:

1. **Multi-series time series charts** for each biomarker group **(Must)**
2. **Clinical range indicators** (normal/abnormal zones) **(Must)**
3. **Interactive tooltips** showing exact values and dates (Optional)
4. **Zoom and pan functionality** for detailed analysis (Optional)
5. **Responsive design** for mobile/desktop **(Must)**
6. **Export functionality** (PNG/PDF) (Optional)

Step 2.3: Dashboard Layout

Create a dashboard with these sections:

- **Header:** Patient overview and date range selector
- **Main Panel:** Large time series visualization
- **Side Panel:** Individual biomarker trend cards
- **Footer:** Clinical interpretations and recommendations



Part 3: Deployment & Testing

Step 3.1: Deployment Options

Replit Deployment:

1. Create new Repl with HTML/CSS/JS template
2. Upload your visualization files
3. Configure custom domain (optional)
4. Share public URL

Vercel Deployment:

1. Connect GitHub repository
2. Deploy with automatic builds
3. Custom domain available

Heroku Deployment (Flask app):

Step 3.2: Quality Assurance Testing

Test the following scenarios:

1. **Data Loading:** Verify all biomarkers load correctly
2. **Visualization:** Check all charts render properly
3. **Interactivity:** Test zoom, hover, and filtering
4. **Responsiveness:** Check mobile and desktop layouts
5. **Performance:** Ensure fast loading times



Part 4: Documentation & Presentation

Step 4.1: Create Documentation

Write README.md covering:

- Project overview and objectives
- Technology stack used
- Installation/setup instructions
- Features and functionality
- Clinical interpretation guide
- Future enhancement opportunities

Step 4.2: Prepare Demo

Create a 5-minute demo covering:

1. **Problem Statement:** Why biomarker tracking matters
2. **Solution Overview:** Dashboard features and benefits
3. **Technical Implementation:** Tools and approaches used
4. **Clinical Insights:** What the data reveals
5. **Next Steps:** Scalability and enhancement plans

Required Deliverables

1. Hosted Dashboard URL

- Working demonstration with sample data
- Interactive time series visualizations
- Mobile-responsive design

2. GitHub Repository



Include:

- Complete source code
- Data processing scripts
- Deployment configuration
- Comprehensive [README.md](#)

3. Technical Documentation

- Architecture decisions log
- Clinical ranges reference
- Performance benchmarks
- Security considerations

4. Demo Video/Presentation

- 5-minute walkthrough
- Problem-solution narrative
- Technical highlights
- Business value proposition



Evaluation Criteria

Technical Implementation (40%)

- **Data Processing:** Clean extraction and transformation (10%)
- **Visualization Quality:** Clear, accurate time series charts (15%)
- **Code Quality:** Well-structured, documented code (10%)
- **Deployment Success:** Working hosted application (5%)

Healthcare Domain Knowledge (25%)

- **Clinical Accuracy:** Correct biomarker ranges and interpretations (15%)
- **User Experience:** Intuitive design for healthcare professionals (10%)

Startup Mindset (20%)

- **MVP Focus:** Essential features prioritized (10%)
- **Rapid Development:** Efficient use of AI tools (5%)
- **Problem-Solution Fit:** Clear value proposition (5%)

Communication (15%)

- **Documentation Quality:** Clear setup and usage instructions (8%)
- **Demo Effectiveness:** Compelling presentation of solution (7%)

Essential Resources & Links

Development Tools

- **Cursor IDE:** <https://cursor.sh/>
- **V0 by Vercel:** <https://v0.dev/>
- **Replit:** <https://replit.com/>
- **Chart.js:** <https://www.chartjs.org/>



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- **D3.js:** <https://d3js.org/>
- **Plotly.js:** <https://plotly.com/javascript/>

Healthcare Data Resources

- **Clinical Ranges Reference:** <https://www.mountsinai.org/health-library/tests>
- **Biomarker Database:** <https://markerdb.ca/>

Hosting Platforms

- **Vercel:** <https://vercel.com/>
- **Netlify:** <https://netlify.com/>
- **Heroku:** <https://heroku.com/>
- **GitHub Pages:** <https://pages.github.com/>

Learning Resources

- **Y-Combinator Startup School:** <https://www.startupschool.org/>
- **Healthcare Data Visualization:** <https://academic.oup.com/nar/>
- **Time Series Analysis:** <https://plotly.com/javascript/time-series/>



Getting Started Checklist

- ☐ Set up development environment (Cursor/Replit/V0)
- ☐ Download sample health reports
- ☐ Create data extraction script
- ☐ Build time series visualization
- ☐ Deploy to hosting platform
- ☐ Write documentation
- ☐ Record demo presentation
- ☐ Submit deliverables

Time Management Tip: Focus on MVP functionality first. Build core time series visualization before adding advanced features. Remember: "Perfect is the enemy of good" - ship a working solution and iterate.

Success Indicator: A healthcare professional should be able to understand a patient's biomarker trends at a glance and identify areas requiring attention.